

# Changwei Yao

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## EDUCATION

### Carnegie Mellon University

M.S. in Information Systems

Advisor: Marios Savvides, Ji Zhang

Pittsburgh, PA

Aug. 2024 - May. 2026 (expected)

### Hunan University

B.E. in Information Security (1/41)

Advisor: Wenqiang Jin

Changsha, China

Sep. 2019 - Jun. 2023

## SKILLS & COURSEWORK

### Operating System

Linux (Ubuntu and Arch), Windows

### Programming Languages

Python, C++, MATLAB, JavaScript, L<sup>A</sup>T<sub>E</sub>X

### Physical Robots

LBR iiwa7, ARX, UR5, myCobot

### Common Libraries

PyTorch, DM\_Control, Isaccgym, Numpy/Scipy, Pandas

### Cousework

Linear Algebra, Calculus, Probability, Robot Learning, Learning for 3D Vision

## PUBLICATION

- **Changwei Yao**, Xinzi Liu, Chen Li, Marios Savvides. "Reward Evolution with Graph-of-Thoughts: A Bi-Level Language Model Framework for Reinforcement Learning." *2026 IEEE International Conference on Robotics and Automatiao (ICRA)*. (under review) [\[paper\]](#)
- Qingyang Wang\*, **Changwei Yao\***, Xingang Liu\*, Zikai Ouyang, Junwei Liu, Haibo Lu, Wei Zhang. "MinInter: Minimizing Trajectory Interpolation During Data Augmentation for Imitation Learning." *2026 IEEE International Conference on Robotics and Automatiao (ICRA)*. (under review)
- Shibo Zhao, Guofei Chen, Honghao Zhu, Zhiheng Li, **Changwei Yao**, Nader Zantout, Seungchan Kim, Wenshan Wang, Ji Zhang, Sebastian Scherer. "SuperMap: Spatio Temporal Semantic SLAM Enabling Robots to Understand Evolving World in Real Time." [\[demo\]](#)
- Zejun Xu, Wenqiang Jin, **Changwei Yao**, et al. "EM-Rhythm: An Authentication Method for Heterogeneous IoT Devices." *ACM Transactions on Sensor Networks* (2024). [\[paper\]](#)
- Miaomiao Yang\*, **Changwei Yao\***, Shijin Yan\*. "Age Estimation Based on Graph Convolutional Networks and Multi-head Attention Mechanisms." *2023 IEEE 6th International Conference on Information Systems and Computer Aided Education (ICISCAE)*. IEEE, 2023.

## EXPERIENCE

### CyLab Biometrics Center

Part-time Research Assistant. Supervised by Prof. Marios Savvides

Sep. 2024 - Now

CMU, PA

- Developed a chat AI agent for navigation at Dubai Airport.
- Enabled LLM to decompose manipulation tasks with **Graph of Thoughts**, helping LLM design more effective reward functions for RL agent in simulation.
- Finding a way to align the **affordance map** and language with the first observed frame of the task. Exploring better methods to enable the **low-level controller** to guide actions more effectively within the VLA framework.

### Ji's Lab at Robotics Institute

Part-time Research Assistant. Supervised by Prof. Ji Zhang

Sep. 2024 - Mar. 2025

CMU, PA

- Introduced a unified **scene graph** to track objects over time, which allows a robot to build an abstract metric-semantic representation.
- Leverage open-vocabulary 3D semantics network to support the detection and tracking of both short-term and long-term **dynamic objects**.

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Control &amp; Learning for Robotics and Autonomy Lab</b>                                                                                                                                                                                                                                                                              | Jun. 2023 - Jun. 2024 |
| Full-time Research Assistant. Supervised by Prof. Wei Zhang                                                                                                                                                                                                                                                                              | SUSTech, China        |
| <ul style="list-style-type: none"> <li>Implemented a <b>distillation</b> method for Diffusion Policy, enhancing inference speed for efficient, low-latency visuomotor control in resource-constrained robots.</li> <li>Explored a novel <b>data augmentation</b> method using model-free optimization for imitation learning.</li> </ul> |                       |
| <b>IoT Sensing and Security Lab</b>                                                                                                                                                                                                                                                                                                      | Oct. 2021 - Aug. 2022 |
| Part-time Research Assistant. Supervised by Prof. Wenqiang Jin                                                                                                                                                                                                                                                                           | HNU, China            |
| <ul style="list-style-type: none"> <li>Researched machine learning, signal processing, and their application in IoT security.</li> <li>Designed a system for heterogeneous IoT device security using electromagnetic signals.</li> </ul>                                                                                                 |                       |

PROJECTS

|                                                                                                                                                                                                                                                                                                                                                              |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Deep Reinforcement Learning for Minecraft</b>                                                                                                                                                                                                                                                                                                             | Apr. 2023 - May. 2023 |
| Personal Project.                                                                                                                                                                                                                                                                                                                                            | Changsha, China       |
| <ul style="list-style-type: none"> <li><b>MineRL Integration:</b> Encapsulated the Minecraft environment within the MineRL library.</li> <li><b>AI Agent Training:</b> Trained a tree-chopping AI agent using Deep Q-Learning to learn from demonstrations and achieve automated task execution.</li> </ul>                                                  |                       |
| <b>Computer Vision Somatic Analysis Method System</b>                                                                                                                                                                                                                                                                                                        | Mar. 2022 - May. 2022 |
| Computer Software Copyright. Authorized by China Copyright Protection Center                                                                                                                                                                                                                                                                                 | Changsha, China       |
| <ul style="list-style-type: none"> <li><b>Software Development:</b> Developed an IoT software using Heartbeat technology for resource takeover, emphasizing multi-threading and dynamics.</li> <li><b>Diverse Functionality:</b> Implemented numerous functions, including Analysis Method View &amp; Management, Posture Analysis Matching, etc.</li> </ul> |                       |

HONORS & AWARDS

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|---------------------------------|-----------------------------------------------------------|
| <b>Second Place Award (20%)</b> | National College Student Information Security Competition |
| <b>Meritorious Winner (10%)</b> | COMAP's Mathematical Contest in Modeling                  |
| <b>Second Place Award (15%)</b> | MathorCup College Mathematical Modeling Contest           |
| <b>Merit Award (5%)</b>         | Electronic Systems Engineering Skills Competition         |